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INTERIM REVISION OF
Fed. Spec. W-F-421b
September 24, 1956

# INTERIM FEDERAL SPECIFICATION

# FLASHLIGHT, ELECTRIC, HAND

This Interim Federal Specification was developed by Standardization Division, Federal Supply Service, General Services Administration, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification W-F-421b, dated September 24, 1956.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers prefocused flashlights, of tubular construction, for general and industrial uses.

1.2.1 Types, classes, styles, and subtypes.

Types:

I. Standard service.
II. Rough Service.
III. Rough service, watertight.
IV. Rough service, watertight and explosion-proof.

Classes:

A. 2-Cell.
B. 3-Cell.

Styles:

1. With spot lens.
2. With diffusion lens.
3. With spot lens and spare bulb.
4. With spot lens, diffusion lens and a spare bulb.

Subtypes:

E. Without holding magnet.
F. With permanent attached magnet.
G. With removable magnet.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

L-P-396 - Plastic Molding Material and Plastic Extrusion Material, Polystyrene.
L-P-1183 - Plastic Molding Material. Acrylonetrile-Butadiene-Styrene (ABS), Rigid.
W-B-101 - Battery,Dry.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-cleated.
PPP-B-601 - Boxes, Wood, Cleated-plywood.
PPP-B-621 - Boxes, Wood, Nailed And Lock-corner.



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PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-wall.
PPP-B-665 - Boxes; Paperboard, Metal Stayed (including Stay Material).
PPP-B-676 - Boxes, Setup.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specification and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 - Preservation, Methods of.

Military Standards:

MIL-STD-105 - Sampling procedures and tables for inspection by attributes.
MIL-STD-129 - Marking for shipment and storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

United States Department of Interior:

Bureau of Mines, Schedule 10C - Electric Mine Lamps.

(Application for copies should be addressed to the Publication Distribution Section, Bureau of Mines, 4800 Forbes Avenue, Pittsburgh, PA. 15212.)

Underwriters' Laboratories, Inc., (UL) Standard:

No. 783 - Electric Flashlights and Lanterns for use in Hazardous Locations, Class I, Groups C and D.

(Application for copies should be addressed to the Underwriter's Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 1285 Walt Whitman Road, Melville, Long Island, New York 11746; or 1655 Scott Blvd., Santa Clara, Calif. 95050.)

3. REQUIREMENTS

3.1 General. The flashlights furnished under this specification shall be of tubular design consisting of battery housing, reflector, lens, retaining ring, lamp, lampholders, switch and additional requirements as specified herein. The flashlights shall be designed to operate on D type dry cell batteries that are in accordance with Federal Specification W-B-101. Batteries shall not be furnished.



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3.1.2 Preproduction sample. When specified in the invitation for bids preproduction samples of the flashlights shall be submitted to the contracting officer for test and approval (see 6.2).

3.2. Types.

3.2.1 Type I. Standard service flashlights shall be a commercial type designed for general usage and shall meet all requirements of this specification that are applicable to the type I.

3.2.2 Type II. These flashlights shall be a commercial type designed for rough service and shall meet all the requirements of a type I flashlight plus the drop test (see 4.3.16).

3.2.3 Type III. These flashlights shall be watertight and designed for rough service usage. Type III flashlights shall meet or exceed all the requirements of this specification with the exception of the explosion proof requirements.

3.2.4 Type IV. These flashlights shall meet or exceed all requirements of this specification. The flashlight shall be waterproof, explosion proof, and shall meet or exceed the requirements of the Bureau of Mines Schedule 10C, and UL 783. In the event of any conflict between this specification, UL 783, and Schedule 10C, the requirements of the Bureau of Mines Schedule 10C shall govern.

3.2.4.1 Explosion proof. Each explosion proof flashlight shall have a label molded on the side of the case indicating that the flashlight is approved by the Bureau of Mines for safety in methane air mixtures, and shall include the Bureau of Mines approval number and name of supplier. In addition, the following information shall be included on the same label or on a separate label: FOR USE IN THE FOLLOWING EXPLOSIVE ATMOSPHERES ONLY: Gasoline, Petroleum, Naptha, Alcohol, Acetone, Lacquer Solvent Vapors and Natural Gas.

3.3 Material.

3.3.1 Material general. The materials throughout shall be suitable in every respect for the intended purpose. The flashlights and the component parts shall evidence no indication of damage, penetration of protective coating or corrosion, when subjected to the tests specified in section 4. Dissimilar metals used in contact with each other shall be carefully selected to prevent electrolytic corrosion.

3.3.2 Case material for types I and II. Unless otherwise specified (see 6.2), case, caps and lens rings may be metal or plastic or a combination thereof. The metal case flashlights shall be either copper coated steel or aluminum, with a chrome plated or enamel finish, that meets the requirements of this specification. The plastic case flashlights shall be constructed of materials of known and proven quality for the intended purpose. Unless otherwise specified (see 6.2), with the exception of colors olive drab and magenda, the battery housing colors of the types I and II shall be the manufacturers option. Switch contacts for the type I flashlight may be of a suitable material such as brass coated steel provided the materials used have suitable electrical conductivity characteristics and meet all requirements of this specification. Switch contacts for the type II flashlight shall be of commercial grade or better spring brass, phosphor bronze or beryllium copper. End or base cap spring of the types I and II may be of a suitable material such as copper coated steel provided the materials used have suitable electrical conductivity and meet all the requirement of this specification.

3.3.3 Case materials for types III and IV. The case, caps, lens rings, and storage compartments for spare lamps and lens shall be of molded plastic that meets or exceeds the mechanical and physical requirements given in table I, type VI of L-P-1183. Any conflict between L-P-1183 and this specification, the requirements of this specification shall govern. The above reference to L-P-1183 is used to establish a minimum standard for plastic used in the manufacture of flashlights types III and IV. Switch contacts shall be of commercial grade or better, spring brass, phosphor bronze or beryllium copper. End base cap spring may be of a suitable material such as copper coated steel provided the materials used have suitable electrical conductivity characteristics and meets all applicable requirements of this specification. Unless otherwise specified (see 6.2), the battery case colors for the type III shall be olive drab number 14087 and the battery case color for the type IV shall be magenta, number 17142. The colors shall be in accordance with Fed. Std. No. 595. Colors of the end cap and lens ring shall be the manufacturers option.



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3.4 Subtypes.

3.4.1 Subtype F. Subtype F flashlights shall be furnished with a permanent steel magnet, permanently attached to the battery case of the flashlight. The magnet may be part of the switch or mounted elsewhere on the battery case. The flashlight shall meet the requirements of section 4 (see 4.3.1 and 4.3.8).

3.4.2 Subtype G. Flashlights shall be furnished with a detachable steel spring band not over two inches long, open at one side. A permanent steel magnet shall be permanently attached to the outside of this band. The band shall be designed to fit over the battery case of a fully assembled flashlight with enough friction fit tension to hold the flashlight in position. The band shall meet test requirements specified in section 4 (see 4.3.2).

3.5 Construction.

3.5.1 Lens. The lens shall be formed from clear plastic that will not discolor, be affected by salt water, oil or gases. The lens shall meet or exceed the mechanical and physical requirements of L-P-396 type III class 2. Glass lens may be used in type I and II flashlights providing it meets the test requirements of section 4. Any conflict between this specification and L-P-396, the provisions of this specification shall govern.

3.5.1.1 Spot lens. The spot lens when assembled to the flashlight shall project a concentrated beam of light showing on the screen, the plane of which is perpendicular to the optical axis of the flashlight at a distance of five (5) feet from the lens, shall not exceed 9 inches in diameter.

3.5.1.2 Diffussion lens. The diffussion lens when assembled to the flashlight shall project a beam of light such that concentrated the pattern shows on the screen, the plane of which is perpendicular to the optical axis of the flashlight, at a distance of 4 feet from the lens, shall be between 60 and 80 inches in diameter. The light pattern shall have uniform distribution of light except at the center where some concentration of light is desired.

3.5.1.3 Lens transmission. The light transmission of lens shall be at least 83 percent.

3.5.1.4 Lens diameter. The diameter of the lens shall not be less than 1-1/2 inches.

3.5.2 Reflector. The contour of the reflector shall be essentially parabolic. The reflecting surface shall be vacuum or electroplated and polished to assure a coefficient of reflection that will meet the requirements of this specification. The reflector shall be lacquered on the inside only. The lacquer shall not react chemically with either the gaskets or the reflector material and shall have sufficient rigidity to prevent distortion when the lamp and batteries are in place. The reflecting surface shall be smooth and free from bubbles or pits.

3.5.3 Battery contact spring. The length and design of the rear contact spring shall accommodate the required number of batteries of any length between 2-5/16 inches to 2-13/32 inches. The spring shall compress a minimum of 3/8 inch and exert a force of not less than 2 pounds.

3.5.4 Roll. The flashlight fully assembled, with batteries, shall be constructed so that when placed on a hard stationary flat surface, inclined $5^{\circ}$ for the type I and $10^{\circ}$ for types II, III, and IV to the horizontal line, the flashlight shall not slip or slide and shall not roll more than 4-1/2 inches (see 4.3.5).

3.5.5 Suspension holder. The suspension holders shall be capable of supporting without distortion a weight of 15 pounds for the two cell flashlights and 25 pounds for the three cell flashlights. The loop of the holder shall, unless removed by force by the operator, be designed to seat firmly into the end cap or lay flat against the battery case. The case mounted holders shall be located opposite the switch close to the end cap. Enough clearnace between the switch and the holder shall be maintained to accommodate the insertion of the magnet band. Unless specified (see 6.2), type I flashlights shall not be required to have a suspension holder. Suspension holders of the type IV flashlights shall be made of a suitable copper alloy.

3.5.6 Lamp and lampholders. The lamps furnished with the flashlights shall be prefocused minature flange base, tungsten filament, single contact of the commercial type PR2 or PR3. The lamp shall be solidly retained in place in the lampholder. The type D dry cells shall be held in place by a helical spring or other acceptable means. When requested by the contracting officer (see 6.2), lamps PR6 or PR7 will be furnished in lieu of the standard issue.



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3.5.7 Switch. The switch shall be mounted on the side of the case and shall be a slide type which will be solidly retained in position when placed in either the "ON" or "OFF" positions. Switches shall resist accidental movement to the "ON" position. The switch slide or button shall be designed to permit easy manipulation with one finger of the same hand holding the flashlight. The design and construction of the switch mechanism shall be such that it will operate in a satisfactory manner, for 35,000 continuous cycles at a minimum rate of 15 cycles per minute (see 4.3.8). The flashlight when completely assembled, fresh dry cells in place, and the switch in the "OFF" position, it shall be impossible to energize the lamp by bridging any two points on the exposed surface of the electrical conductor. A flashing position shall be required on all types III and IV flashlights. When requested (see 6.2), a flashing position shall be furnished on types I and II flashlights.

3.5.8 Storage compartment. Storage compartment, when requested (see 1.2.1 styles), shall be mandatory on the types III and IV flashlights only. The storage compartment shall be provided in the end caps and shall be of the same material as used in the case. The storage area shall be so designed that the lamp and lens, are readily accessible, fit securely in the compartment, and will not become broken or misplaced when the flashlight is subjected to tests in section 4.

3.5.9 The dimensions of the flashlight shall be of the values given in table I (see 4.9).

TABLE I.

| Type class Style | Lamp size | Maximum Overall dimensions in inches excluding switch | | Illumination Candle Power | | | |
|---|---|---|---|---|---|---|---|
| | | | | Center and points on 4 inch circle | | 8 inch Circle 4 points | |
| | | Length | Diameter | Minimum | Minimum Average | Minimum | Minimum Average |
| I-A-1 | PR2 | 7-1/2 | 2-1/4 | 600 | 800 | 300 | 350 |
| II-A-1 | PR2 | 7-1/2 | 2-1/2 | 600 | 800 | 300 | 350 |
| III-A-3 | PR2 | 8 | 2-1/2 | 600 | 800 | 300 | 350 |
| IV-A-4 | PR2 | 9 | 2-1/2 | 600 | 800 | 300 | 350 |
| I-B-1 | PR3 | 10-1/2 | 2-1/2 | 1200 | 1500 | 500 | 550 |
| II-B-1 | PR3 | 10-1/2 | 2-1/2 | 1200 | 1500 | 500 | 550 |
| III-B-3 | PR3 | 11 | 2-1/2 | 1200 | 1500 | 500 | 550 |
| IV-B-4 | PR3 | 12 | 2-1/2 | 1200 | 1500 | 500 | 550 |
| | | | | | | | |
| | | | | | | | |

3.5.10 Illumination. When the flashlight beam is projected onto a screen having a 4-inch circle and a 8-inch circle, the light readings taken shall meet or exceed the requirements of table I.(see 4.3.10).

3.5.11 Material stability There shall be no visual or physical change in plastic flashlights types II, III, and IV when exposed to water, oil and gases (see 4.3.11).

3.5.12 Threads. Matching threads on the caps, rings and case shall be of the design and fit that may be screwed tight using a torque not to exceed 25 inch pounds. Threads of the types I and II caps, rings and case, shall not strip or ride over those of mating parts when a torque of 40 inch pounds ± 3 inch pounds is applied. Threads of the types III and IV caps, rings, and case, shall not strip or ride over those of mating parts when a torque of 75 inch pounds ± 3 inch pounds is applied. For types I and II (see 4.3.12), for types III and IV (see 4.3.13).



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3.5.13 Gaskets. Gaskets used in types III and IV flashlights shall be provided at any joint through which moisture could possibly pass into the interior of the flashlight. The gaskets may be of the manufacturer's option but shall be resistant to oil and water and shall meet the test requirements of this specification (see 4.3.13).

3.5.14 Case. For secure gripping the outside surface of the battery case shall be fluted, ribbed or knurled to a minimum depth of 0.025 inches. Types I and II flashlights may be furnished in a one piece body construction or with threaded end caps providing, they meet the requirements of this specification (see 3.5.15). Types III and IV flashlights shall be furnished with end caps.

3.5.15 Caps and lens rings. Means shall be provided through the use of threaded caps and retaining rings for securing the lens and batteries in place. Matching threads of all caps, retaining rings, and cases, shall have a minimum of 1/16 inch seating clearance before the beginning of the first thread. Types I and II flashlights, without end caps, shall have a minimum of two engaged threads for all metal flashlights, and a minimum of three engaged threads for flashlights using plastic threads. Type I and II flashlights with end caps shall have a minimum of one engaged thread for all metal flashlights and a minimum of two engaged threads for flashlights using plastic threads. Type III and IV flashlights shall have a minimum of three engaged threads.

3.5.16 Durability. The flashlights shall be designed and tested to meet the requirement of the user. The type I flashlight shall be tested in accordance with test paragraph 4.3.15. Type II flashlights shall be designed to meet the requirements of a repairman working from a six-foot ladder and shall meet the test requirements 4.3.16. Type III flashlights shall be designed to meet the requirements of a repairman working in water and from a six-foot ladder and shall meet the test requirements of 4.3.14 and 4.13.16. Type IV flashlights shall be designed to meet the requirements of a repairman working in water, from a 6-foot ladder in areas where there is a danger from explosive gases. Therefore, these flashlights shall meet the test requirements of 4.3.14, 4.3.16 and Bureau of Mines Schedule 10C.

3.5.17 Impact. Upon completion of test specified in section 4, types III and IV flashlight shall be visually checked for moisture seepage into the internal parts of the flashlight and all types of flashlights shall be checked for malfunction.

3.5.18 Interchangeability. All corresponding parts, of flashlights of the same type shall be interchangeable.

3.5.19 Cleaning. All parts shall be thoroughly cleaned. Traces of corrosive compound used for cleaning, plating, finishing or machining shall be removed or chemically neutralized to prevent subsequent corrosion.

3.5.20 Workmanship. The flashlights shall be free from defects such as sharp edge, improper plating, painting, discoloration, chipping, cracking of the finish or any visual defects which detract from the appearance or serviceability of the flashlights. There shall be no evidence of loose rivets or screws or any indication of deterioration of any component when subjected to tests in section 4.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection conditions. Unless otherwise specified herein, all inspections shall be made at room temperature, pressure and humidity.

4.1.2 Test flashlights. Unless otherwise specified (see 6.2), the spot lens, size D batteries, PR2 bulb for two-cell flashlights, PR3 bulb for three-cell flashlights shall be used in the following tests. The tests shall be in sequence starting with 4.3.1.



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4.2 Sampling for acceptance inspection.

4.2.1 Sampling for acceptance.

4.2.1.1 Sampling for group A inspection tests. Sampling for group A inspection shall be in accordance with MIL-STD-105, inspection level II, Acceptance Quality Level (AQL) 2.5 and each sample flashlight shall be subjected to group A tests listed in table II.

4.2.1.2 Sampling group B tests. Sampling in group B tests shall be in accordance with MIL-STD-105, inspection level S-2, AQL 2.5, and each sample flashlight shall be subjected to group B tests specified in table II.

4.2.1.3 Sampling group C tests. Group C tests shall be required at the beginning of each contractual year. Retest of the flashlights shall be required when there is a design change or when requested by the contracting officer. Three complete flashlights of each type, class, style and subtype contracted for shall be subjected to group C tests.

TABLE II.

| Examination and tests | Group A<br>Requirement paragraph | Test paragraph |
|---|---|---|
| General examination | 1.2.1, 3.1<br>3.5.2, 3.5.6, 3.5.8,<br>3.5.9, 3.5.14, 3.5.18,<br>3.5.19, 3.5.20, 3.2.4.1 | Visual and dimensional |
| | Group B | |
| Performance | 3.5.1.1<br>3.5.1.2<br>3.5.15<br>3.5.17 | 4.3.10<br><br>4.3.14<br>Test C for types I and II. Tests B and C for types III and IV. |
| | Group C | |
| Subtype | 3.4 | 4.3.1, 4.3.2 |
| Battery contact spring | 3.5.3 | 4.3.4 |
| Roll | 3.5.4 | 4.3.5 |
| Suspension holder | 3.5.5 | 4.3.6 |
| Impact | 3.5.17 | 4.3.7 |
| Switch | 3.5.7 | 4.3.8, 4.3.8.1 |
| Material stability | 3.5.11 | 4.3.9 |
| Impact | | 4.3.11 |
| Threads | 3.5.12 | 4.3.12 |
| Gasket | 3.5.13 | 4.3.13 |
| Watertightness | | 4.3.14 |
| Durability | 3.5.16 | 4.3.15, 4.3.16 |



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4.2.2 Acceptance inspection.

4.2.2.1 Acceptance group A and B examination and tests. The results of each examination and test shall be compared with the requirements of this specification. Failure to conform to the requirements of this specification for any tests shall be counted as a defect, and the flashlight shall be rejected. If the number of such nonconforming flashlights in any sample exceeds the acceptance number of that sample, the lot represented by the sample shall be rejected.

4.2.2.2 Acceptance group C tests. The result of each test shall be compared with the requirements of this specification. In the event of failure to conform to the requirements of this specification for any group C tests, the lot represented by the sample shall be rejected.

4.3 Inspection and tests.

4.3.1 Permanently attached flashlight magnets. The flashlight shall be mounted to a metal surface by magnetism in such a manner that the longitudinal axis of the flashlight shall be at a 90° angle to the floor. Enough weight shall be added to the flashlight so that the flashlight will weigh a minimum of 6 pounds. The flashlight shall remain in this position for 16 hours. Any movement of the flashlight shall constitute failure of this test (see 3.4.1).

4.3.2 Spring band test. The magnet shall be permanently attached to the band. The spring band shall be mounted to a metal surface by magnetism in such a manner that the longitudinal axis of the flashlight when inserted into the band will be at a 90° angle to the floor. The flashlight shall hang supported by the band alone. During the following test, the band shall remain firmly fixed to the metal surface by magnetism. A round metal bar 2-inches in diameter shall be passed through the spring tension arc, seated for not less than five seconds, then withdrawn, constituting one cycle. The test shall consist of 100 cycles. Then a 1-1/2-inch diameter smooth metal bar weighing not less than six pounds shall be inserted and seated in the spring band. The 6 pound bar shall be held in this position by spring tension for a period not less than 16 hours. Any evidence of slippage or movement of the bar or magnet shall constitute failure of this test. The magnet and band shall remain in the same position throughout the tests (see3.4.2).

4.3.3 Lens. Spot and diffusion lenses shall be tested to determine compliance with requirements in paragraphs 3.5.1.

4.3.4 Battery contact spring. The rear battery contact spring shall be tested as follows:

(a) Determine available length of battery compartment.

(b) Subtract 4-5/8 inches (total minimum length of two Size D batteries).

(c) Set the spring on a flat surface and add a 2-pound weight to the top of the spring. Measure the height of the spring and compare against difference of (a)-(b) above. If the height of the spring (see above) is less than a-b, the flashlight shall have failed this test.

4.3.5 Roll. The flashlight shall be positioned at three different points on its's diameter, 90° apart. With the flashlight positioned under maximum roll conditions on the flat surface inclined 5° for the type I and 10° for the types II, III, and IV, the flashlight shall not roll more than 4-1/2 inches (see 3.5.4).

4.3.6 Suspension ring. The suspension ring test shall use a 15-pound weight for the class A flashlight and a 25-pound weight for the class B flashlights. With the flashlights in a vertical position, base down, the required weight shall be suspended from the ring for a period of 1 minute. After this test the ring shall show no signs of distortion and will seat firmly to and recess into the end cap (see 3.5.5) or to the side of the case.

4.3.7 Impact. Plastic flashlights types III, and IV shall be subjected to a temperature of -30°F. The types I and II flashlights shall be subjected to a temperature of -20°F. ± 2°F. The flashlights shall be subjected to the temperatures indicated above for 2 hours and shall then be subjected to a 12-inch pound impact using a 1-pound steel ball. The impact shall be applied (a) to the outside of the base tube at a point on the side opposite the switch; (b) to the lens ring; (c) to the bottom cap. Failure to operate normally, or any chipping or cracking shall constitute failure of this test.



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4.3.8 Switch. The switch mechanism of the flashlight shall be tested by operating the button for 35,000 continuous cycles at a minimum rate of 15 cycles per minute, not to exceed 30 cycles per minute. A cycle shall consist of movements from "OFF" position to full "ON" position and back to "OFF" position. The swtich shall be operated under normal electrical load conditions. The lamp and batteries shall be renewed as often as required to assure that the switch mechanism is operating under normal load throughout the 35,000 cycles (see 3.5.7). Failure of this switch to complete 35,000 cycles shall constitute failure of this test.

4.3.8.1 Switch leakage. With the lamp and reflector removed, insert batteries into the flashlight, connect a voltmeter (Simpson Model 260 or equal) across the switch and battery or the switch and battery contacts as appropriate, in such a way as to read the battery voltage through the switch. With the switch in the "OFF" position read the voltage. Any distinguishable deflection of the meter hand when the meter is set in the voltage range nearest the battery voltage shall constitute failure of this test.

4.3.9 Dimensional and material stability. The following test shall be made on completely assembled flashlights, including batteries, or dummies of equal weight and size. The flashlight shall be placed on a horizontal surface and subjected to a temperature of + 150°F. ± 5°F. dry heat for 16 hours, followed by a temperature of + 110°F. ± 2°F. at 90 percent relative humidity for 6 hours. The flashlights shall then be compared with untested flashlights for dimensonal stability, crazing of surface, or any other visual defects. The parts of the tested samples shall be interchangeable with corresponding parts of untested samples. The dimensions and illumination shall be checked to determine that they conform to the requirements of table I. Nonconformance to applicable dimensions and illumination appearing in table I shall constitute failure of this test.

4.3.10 Illumination. The flashlights illumination readings shall be taken from in front of the screen at the designated points. The flashlight shall be arranged so that its beam is projected on a screen, the plane of which is perpendicular to the optical axis of the flashlight with ten (10) feet between the lens and the screen as described below. For test purposes, the geometric center of the projected beam of light may be considered the point at which the optical axis of the flashlight pierces the test plane. Concentric circles, one of eight (8)-inch diameter and one of four (4)-inch diameter, shall be drawn on the screen. The intersection of these circles with their vertical and horizontal diameters shall be indicated. The intensity of the light falling on the screen at the eight (8)-points of intersection and at the center of the circles shall be read using a light sensitive element having a diameter of 1.00 + 0.05 inches, centered on the points. The flashlight shall be turned on for 15 second intervals, and the reading for a point shall be taken at the end of the time interval. The flashlight will be turned off for 30 seconds between readings. The center point shall be read first, followed by the points on the inner circle and the points on the outer circle.

4.3.10.1 Conversion formula: Candle power - 100 X meter reading in foot candles. The light measuring equipment shall have a known accuracy of ± 5 percent or better. The illumination (candle power) shall be equal to or better than that specified in table I. Manufacturers reading of each flashlight tested shall be noted and shall then be compared with recorded reading taken after the drop tests. The average of the two readings (9-points) shall be within 10 percent. Nonconformance to the above shall constitute failure of this test.

4.3.11 Oil resistance. The test shall be conducted on plastic flashlights types II, III, and IV. Types III and IV flashlights shall be fully charged prior to the test. The sample flashlights shall be submerged for 15 minutes in a mineral oil bath maintained at 125°F. The flashlights shall be carefully examined before and after the immersion bath. Any change in the finish, color, dimensions or any other physical change in the flashlight will constitute failure of this test. Light intensity of types III and IV shall be the same before and immediately after the oil bath. The flashlight outside surface may be wiped clean before checking light intensity. The types III and IV lights shall be disassembled all parts shall be carefully checked to determine that there are no traces of oil on the internal parts of the flashlight. Failure of the types III and IV to operate or any traces of oil on the internal parts shall constitute failure of this test.

4.3.12 Thread. The following thread test is for types I and II flashlights only. Using a minimum torque of 10 inch pounds screw the cap and lens rings up tight (see 3.5.12), and then remove both caps and ring, this will be designated as one cycle. The test shall require that 100 such cycles be performed on the three flashlights specified. At the end of the test period, the flashlights shall be reassembled using the torque of 40-inch pounds ± 3-inch pounds (see 3.4.12). Remove caps and rings, physically examine threads for any sharp edges or weakness.



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4.3.13 Gasket and thread. This test is for types III and IV, flashlights only. Threaded lens rings and caps, of a fully assembled flashlight including batteries, shall be removed and reassembled to a water tight condition, using the inch pound torque established by the manufacturer, not to exceed 15-inch pounds. This shall be designated as one test cycle. The test shall consist of 100 test cycles. The applied torque in each cycle shall be within 1/4-inch pound of the manufacturers established torque. In each cycle the flashlight shall remain in a tightened condition for a minimum of two minutes. At the completion of the above test the flashlights shall be assembled and a torque of 75-inch pounds shall be applied to each of the threaded parts. The flashlight and parts shall then be examined for compliance with requirement 3.5.12. The flashlight shall then be reassembled using inch torque pounds established by the manufacturer as stated above. The flashlight shall remain in this test tightened condition and shall be used as samples in the water tightness test (see 4.3.14).

4.3.14 Watertightness. For types III and IV only. The gasket and thread tests (see 4.3.13), shall precede this test. The tests shall be in three parts designated A, B, and C.:

(A.) The flashlight shall be submerged for a period of one hour to the depth of 3 feet below the surface of a 10 percent solution of sodium chloride at a room temperature of $75^{\circ} \pm 2^{\circ}F$.

(B.) The (B) test shall include that quantity of flashlights required under the sampling plan. Flashlights under this plan shall be charged and a torque, not exceeding 15-inch pounds, shall be applied to cap and lens ring. The flashlight shall then be submerged for a period of one hour to the depth of three-feet below the surface of clear water at the room temperature of $75^{\circ} \pm 2^{\circ}F$.

(Upon the completion of the above tests A and B the flashlights shall be submitted to test C.)

(C.) Test C requires that the flashlight be shaken vigorously for a minimum of 30 seconds, the switch shall then be operated 15 times from full "OFF" position to full "ON" position. The tested flashlight shall then be compared with a nontested flashlight (freshly charged) of the same type, class and style used for this test. There shall be no visual or physical difference in the operation of the flashlights.

4.3.15 Type I, durability drop. This test shall follow the thread test (see 4.3.12). The same flashlights shall be used in both tests. The assembled flashlight, complete with dry cells, lamps and lens, shall be dropped in free fall through a distance of 8-feet on the center of a bed of loose dry sand. The sand shall be at least two-feet by two-feet by six-inches deep and the exposed surface shall be smoothed to a horizontal plane prior to each drop. Each flashlight shall be dropped so that it strikes ten times in an approximate horizontal position, ten times in a vertical position upon the lens and ten times in a vertical position upon the base of the flashlight. Tightening of threaded parts shall not be made once the test has started. At the completion of this test flashlights shall be tightened using a 40-inch torque pounds and shall meet the requirements specified in 4.3.12. The flashlights shall then be shaken vigorously for 30 seconds and shall be submitted to 15 cycles test operation (see 4.3.8). Upon completion of the above tests, the flashlights shall be tested in accordance with 4.3.10. A new lens may be used if abrasive action of the sand has caused scratches that prevent the type I flashlight from passing, the test specified in 4.3.10. The breaking, cracking or chipping of the lens shall be regarded as failure of this test.

4.3.16 Rough service durability. Tests 4.3.13 and 4.3.14 for types III and IV and test 4.3.12 for the type II shall preceed the following tests. The assembled flashlights complete with batteries, lamp and lens, shall be dropped in a free fall through a distance of 7-foot on to a concrete floor. The flashlights shall be dropped horizonally so that the flashlights will land, twice on the switch, and twice on the side opposite the switch. In the vertical position, twice on the lens and twice on the end caps. Upon completion of the above tests types III and IV shall be submitted to the water tightness test "B" (see 4.3.14.B), for an emersion period of not less than 10 minutes. Types II, III, and IV flashlight shall then be tested in accordance with test C (see 4.3.14c). The flashlight shall remain in the above final test condition and shall immediately be tested in accordance with the requirements of paragraph 4.3.10. Types III and IV flashlights shall then be disassembled, any evidence of leakage shall constitute failure of this test. Chipping of the finish will be acceptable providing the flashlight meets the test requirements of this specification.



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4.3.17 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table II. The sample unit shall be in accordance with MIL-STD-105. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

TABLE II. Classification of preparation for delivery defects

| Examine | Defects |
|---|---|
| Markings (exterior and interior) | Omitted; incorrect; illegible; improper size; location, sequence, or method of application. |
| Materials | Any component missing or damaged. |
| Workmanship | Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container. |

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. Each flashlight shall be individually packaged in accordance with MIL-P-116, method IC-2.

5.1.1.2 Intermediate packaging. Twelve flashlights of like stock number, packaged, as specified in 5.1.1.1, shall be intermediate packaged in a close-fitting box conforming to PPP-B-636, class weather-resistant. Each box shall be closed in accordance with the appendix to PPP-B-636.

5.1.2 Level B.

5.1.2.1 Unit packaging. Each flashlight shall be individually packaged in close-fitting box conforming to PPP-B-566; PPP-B-665 or PPP-B-676. Each box shall be secured to prevent accidental opening.

5.1.2.2 Intermediate packaging. Twelve flashlights of like stock number, packaged, as specified in 5.1.2.1, shall be intermediate packaged in a close-fitting box conforming to PPP-B-566; PPP-B-665; PPP-B-676 or PPP-B-636, class-domestic. Each box shall be secured to prevent accidental opening.

5.1.3 Level C.

5.1.3.1 Unit packaging. Each flashlight shall be packaged in accordance with the manufacturers standard practice providing that this will insure protection for the flashlight during shipment and provides for safe delivery to its destination.

5.1.3.2 Intermediate packaging. Twelve flashlights of like stock number, packaged, as specified in 5.1.3.1, shall be intermediate packaged in accordance with the manufacturers standard practice, providing that this will insure protection for the flashlights during shipment and provides for safe delivery to their destination.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Flashlights of like description, packaged, as specified in 5.1, shall be packed in quantities, as specified (see 6.2), in a close-fitting box conforming to PPP-B-585, class 3; PPP-B-591, class II; PPP-B-601, overseas type; PPP-B-621, class 2 or PPP-B-640, class 2. Closure of boxes shall be in accordance with the applicable box specification or appendix thereto.



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5.2.2 Level B. Twelve intermediate packages (144 flashlights) packaged, as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-636, class weather resistant, grade 275, minimum. Closure shall be in accordance with the appendix to PPP-B-636.

5.2.3 Level C. Twelve intermediate packages (144 flashlights) packaged, as specified in 5.1, shall be packaged in containers that will assure carrier acceptance and safe arrival at destination in compliance with the Uniform Freight Classification Rules and the National Motor Freight Classification Rules.

5.3 Marking. Marking shall be in accordance with 5.3.1 or 5.3.2, as specified (see 6.2).

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, all interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, all interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Flashlights covered by this specification are suitable for use by personnel in the repair and maintenance of equipment. The four types specified herein are intended for the following uses:

Type I. General use.

Type II. General all around use where rough service is a requirement.

Type III. Rough service flashlight designed for general all-around usage including working in and around water.

Type IV. Rough service flashlight designed for use in and around water and where there is danger from explosions of surrounding combustible, gases, dust, or explosives.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following data in procurement documents:

(a) Title, number, and date of this specification.

(b) Type, class, style, and subtype (see 1.2.1).

(c) Color of case, if other than specified (see 3.3.3.2 and 3.3.3).

(d) Responsibility for inspection, if other than specified (see 4.1).

(e) Quantity of unit packages of flashlights required in shipping container(see 5.2.1).

(f) Selection of the applicable level of packaging and packing required (see 5.1 and 5.2).

(g) Marking, if other than specified (see 5.3).

(h) Lamps, if other than specified (see 3.5.6).

(i) If specific material is required, so state (see 3.3.2 and 3.3.3).

(j) Flashing poisition required for the types I and II flashlights so state (see 3.5.7).

(k) If suspension holder is required for the type I so state (see 3.5.5).

(l) Request for preproduction sample (see 3.1.2).

(m) Test lens and bulbs different than specified (see 4.1.2).





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